

From functional analysis to complexity analysis in optimisation

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Abstract

Modern industries provide a large spectrum of complex optimisation problems which, most of the time, cannot be solved exactly. In these cases, we are often called to use a approximated model to balance between the accuracy and the complexity. Choosing the right model requires a great knowledge about different algorithms complexity. This talk will focus on minimizing functions in compact sets. We will start by reviewing some fundamental concepts in functional analysis and make the connection with current research directions in optimisation.

1 Warm up

Consider a function in $f : [0, 1] \rightarrow \mathbb{R}$. We note $f_\star = \min_{x \in [0, 1]} f(x)$ and we fix an value $\epsilon > 0$, our objective is to find $x \in [0, 1]$ such that

$$f(x) - f_\star \leq \epsilon,$$

which we also call it as an ϵ -close solution of f .

We note the class of functions

$$C_i(K) := \{f : [0, 1] \rightarrow \mathbb{R}, f^{(i)} \text{ being } K\text{-Lipschitz}\}$$

1.1 Lipschitz case

Lemma 1.1. *Let's consider $f \in C_0(G)$, the number of iteration required on f to find an ϵ -close solution on f is $\Omega\left(\frac{G}{\epsilon}\right)$*

Proof. By the Lipschitz condition, we know that:

$$\forall x \in \left[x_\star - \frac{\epsilon}{G}, x_\star + \frac{\epsilon}{G}\right], f(x) - f(x_\star) \leq \epsilon$$

Therefore the problem is equivalent of finding an interval of size $\frac{2\epsilon}{G}$ in $[0, 1]$.

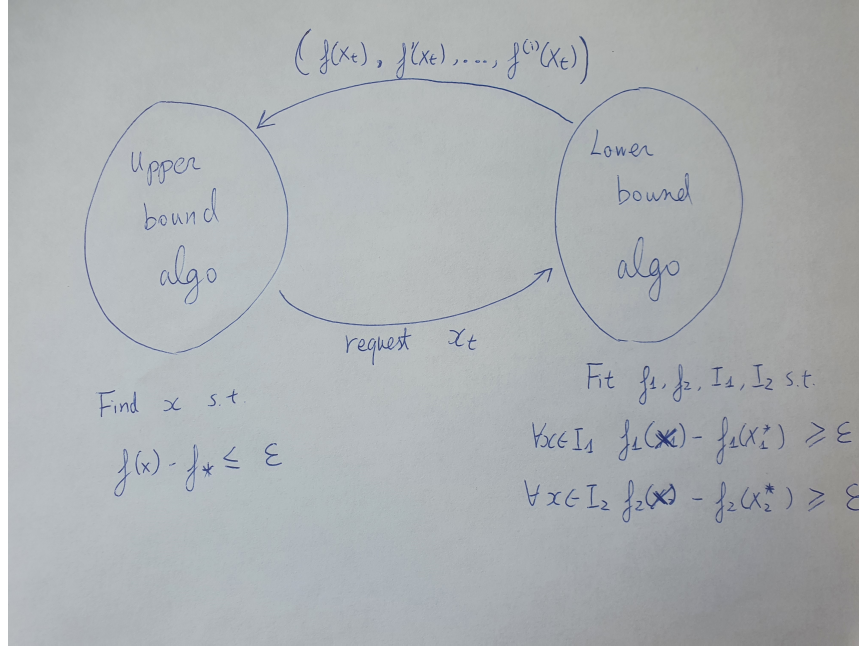


Figure 1: The loop between upper-bound and lower bound at each iteration t

For the upper bound, we can make an algorithm that simply test all the values $x_t = \frac{t\epsilon}{G}$ for t going from 1 to $\lfloor G/\epsilon \rfloor + 1$ and simply returns the minimum value it meets. We are guarantee that one of the $x_{t_0} \in \left[x_* - \frac{\epsilon}{G}, x_* + \frac{\epsilon}{G} \right] \cap [0, 1]$. So the returned value z verifies :

$$f(z) \leq f(x_{t_0}) \leq f(x_*) + \epsilon$$

For the lower bound, we suppose that $N \leq \frac{G}{4\epsilon} - 2$ iterations have been requested and the answer we got has always been :

$$f(x_1) = \dots = f(x_N) = 0$$

For the sake of simplicity, we also consider $x_{N+1} = 0, x_{N+2} = 1$ are known as $f(0) = f(1) = 0$. We consider a permutation σ such that

$$0 = x_{\sigma(1)} \leq x_{\sigma(2)} \leq \dots \leq x_{\sigma(N+2)} = 1$$

There exists i such that : $x_{\sigma(i)+1} - x_{\sigma(i)} \geq \frac{4\epsilon}{G}$.

We can then consider two piece-wise linear functions $f_1, f_2 \in C_0(G)$ such that :

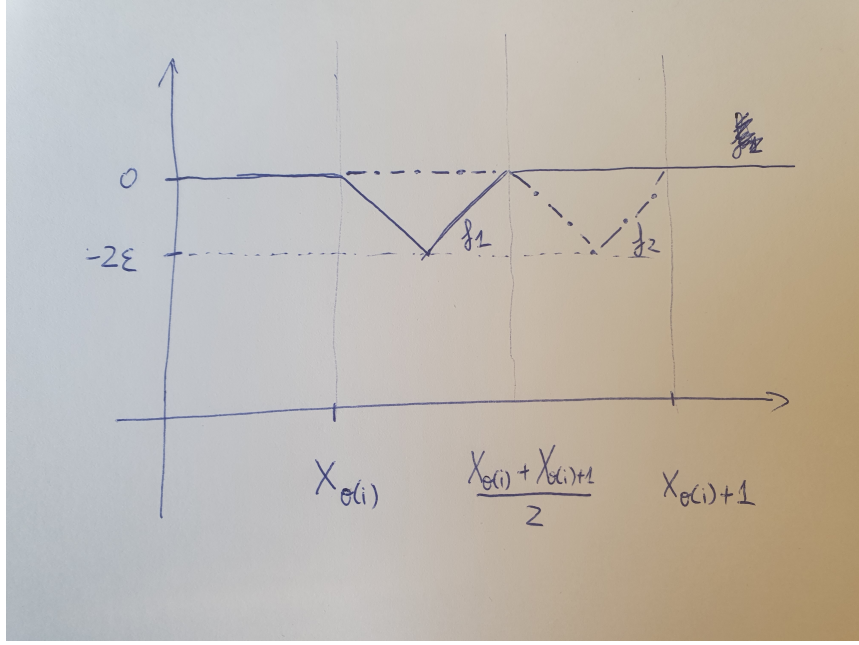


Figure 2: (f_1, f_2) for Lipschitz lower bound

$$\begin{aligned}
 f_1(0) = f_1(x_{\sigma(i)}) = 0, \quad f_1\left(\frac{x_{\sigma(i)} + x_{\sigma(i)+1}}{4}\right) = -2\epsilon, \quad f_1\left(\frac{x_{\sigma(i)} + x_{\sigma(i)+1}}{2}\right) = f_1(1) = 0 \\
 f_2(0) = f_2\left(\frac{x_{\sigma(i)} + x_{\sigma(i)+1}}{2}\right) = 0, \quad f_2\left(\frac{3(x_{\sigma(i)} + x_{\sigma(i)+1})}{4}\right) = -2\epsilon, \quad f_2(x_{\sigma(i)+1}) = f_2(1) = 0
 \end{aligned}$$

We can verify the Lipschitz condition on f_1, f_2 by noticing that :

$$\frac{2\epsilon}{\frac{x_{\sigma(i)+1} - x_{\sigma(i)}}{2}} = \frac{4\epsilon}{x_{\sigma(i)+1} - x_{\sigma(i)}} \leq G$$

Since (f_1, f_2) has separate non-zero sets, no matter what the algorithm returns as z , we can always make sure that one of the two :

$$f_i(z) - f_i(x_\star^i) \geq 2\epsilon > \epsilon.$$

In conclusion, we have a $\lfloor \frac{G}{\epsilon} \rfloor + 1$ upper bound and $\frac{G}{4\epsilon} - 2$ lower bound which differs only up to a constant multiplier if ϵ/G is small enough. $\square$

1.2 Smooth case

Lemma 1.2. *Let's consider $f \in C_1(L)$, the number of iteration required on f to find an ϵ -close solution is $\Omega\left(\sqrt{\frac{L}{\epsilon}}\right)$*

Proof. The argument is similar to the previous lemma. We need to take into consideration that $f'(x_*) = 0$ and then :

$$\begin{aligned} \forall x \in \left[x_* - \sqrt{\frac{\epsilon}{L}}, x_* + \sqrt{\frac{\epsilon}{L}} \right], \quad |f'(x)| &\leq \sqrt{L\epsilon} \\ \forall x \in \left[x_* - \sqrt{\frac{\epsilon}{L}}, x_* + \sqrt{\frac{\epsilon}{L}} \right], \quad f(x) - f(x_*) &\leq \sqrt{L\epsilon} \times \sqrt{\frac{\epsilon}{L}} = \epsilon \end{aligned}$$

The *upper bound* comes also from the choosing $x_i = \frac{i\sqrt{\epsilon}}{\sqrt{L}}$ with i picking from 1 to $\lfloor \sqrt{\frac{L}{\epsilon}} \rfloor + 1$. The previous equation guarantee us there will always be a solution.

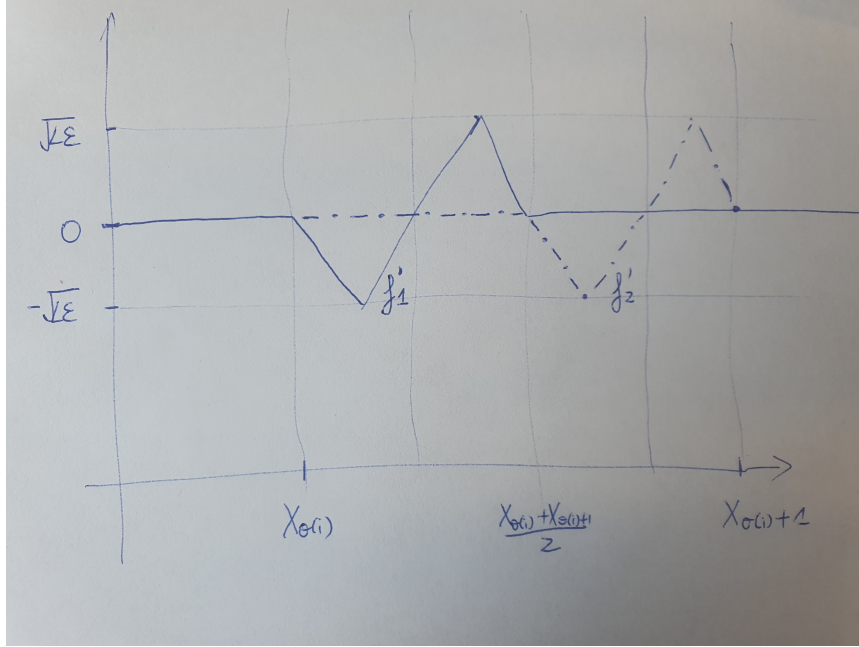


Figure 3: (f'_1, f'_2) for smooth lower bound

The *lower bound* requires a bit more engineering, we will build on f'_1 and f'_2 two piece-wise linear functions. We suppose that $N \leq \frac{1}{8}\sqrt{\frac{L}{\epsilon}} - 2$ iterations have been requested, we consider the permutation σ such that

$$0 = x_{\sigma(1)} \leq x_{\sigma(2)} \leq \dots \leq x_{\sigma(N+2)} = 1$$

There exists i such that : $x_{\sigma(i)+1} - x_{\sigma(i)} \geq 8\sqrt{\frac{\epsilon}{L}}$.

We make linear interpolation

$$\begin{aligned}
f'_1(0) &= f'_1(x_{\sigma(i)}) = 0, & f'_1\left(x_{\sigma(i)} + \frac{x_{\sigma(i)+1} - x_{\sigma(i)}}{8}\right) &= -\sqrt{L\epsilon} \\
f'_1\left(x_{\sigma(i)} + \frac{3(x_{\sigma(i)+1} - x_{\sigma(i)})}{8}\right) &= \sqrt{L\epsilon}, & f'_1\left(\frac{x_{\sigma(i)} + x_{\sigma(i)+1}}{2}\right) &= f'_1(1) = 0 \\
f'_2(0) &= f'_2\left(\frac{x_{\sigma(i)} + x_{\sigma(i)+1}}{2}\right) = 0, & f'_2\left(x_{\sigma(i)} + \frac{5(x_{\sigma(i)+1} - x_{\sigma(i)})}{8}\right) &= -\sqrt{L\epsilon} \\
f'_2\left(x_{\sigma(i)} + \frac{7(x_{\sigma(i)+1} - x_{\sigma(i)})}{8}\right) &= \sqrt{L\epsilon}, & f'_2(1) &= 0
\end{aligned}$$

The smoothness can be verified by :

$$\frac{\sqrt{L\epsilon}}{\frac{x_{\sigma(i)+1} - x_{\sigma(i)}}{8}} = \frac{8\sqrt{L\epsilon}}{x_{\sigma(i)+1} - x_{\sigma(i)}} \leq \frac{8\sqrt{L\epsilon}}{8\sqrt{\frac{\epsilon}{L}}} = L$$

□

2 Convexity and first order methods

2.1 Convex function

Definition 2.1. A function $f \in \mathcal{C}_1$ is convex if for all x, y ,

$$f(x) - f(y) - f'(y)(x - y) \geq 0$$

The left hand side term is called the Bregman divergence of f and is often noted as $D^f(x, y) := f(x) - f(y) - f'(y)(x - y)$. That notion is particularly useful when generalised on higher dimension.

We note $\mathcal{G}$ the set of convex functions. One can notice that by replacing x by x_* , we have :

$$\forall y, \quad f'(y)(y - x_*) \geq f(y) - f(x_*) \geq 0$$

which means that the derivative is negative on $[0, x_*]$ and positive on $[x_*, 1]$. That makes the problem very easy to solve by simply tracking the sign of its derivative.

Lemma 2.1. Let's consider $f \in C_1(L) \cap \mathcal{G}$, the number of iteration required only on f' to find an ϵ -close solution is $\frac{1}{2\log(2)} \log\left(\frac{L}{\epsilon}\right) + \Omega(1)$.

Proof. From the smoothness, we know that :

$$\forall x \in \left[x_* - \sqrt{\frac{\epsilon}{L}}, x_* + \sqrt{\frac{\epsilon}{L}}\right], \quad f(x) - f(x_*) \leq \epsilon$$

By the assumption that $x_* \in (0, 1)$, we know that $f'(0) < 0$ and $f'(1) > 0$.

For the upper bound, we do a binary search on the segment. If we know $g'(a) < 0$ and $g'(b) > 0$, we evaluate on $\frac{a+b}{2}$. If $f'\left(\frac{a+b}{2}\right) > 0$, we iterate on $\left[a, \frac{a+b}{2}\right]$, else we iterate on $\left[\frac{a+b}{2}, b\right]$.

The segment size is reduced by 2 at each iteration, so the algorithm can stop after $\log_2\left(\sqrt{\frac{L}{\epsilon}}\right) + 1 = \frac{1}{2\log(2)} \log\left(\frac{L}{\epsilon}\right) + 1$ iterations.

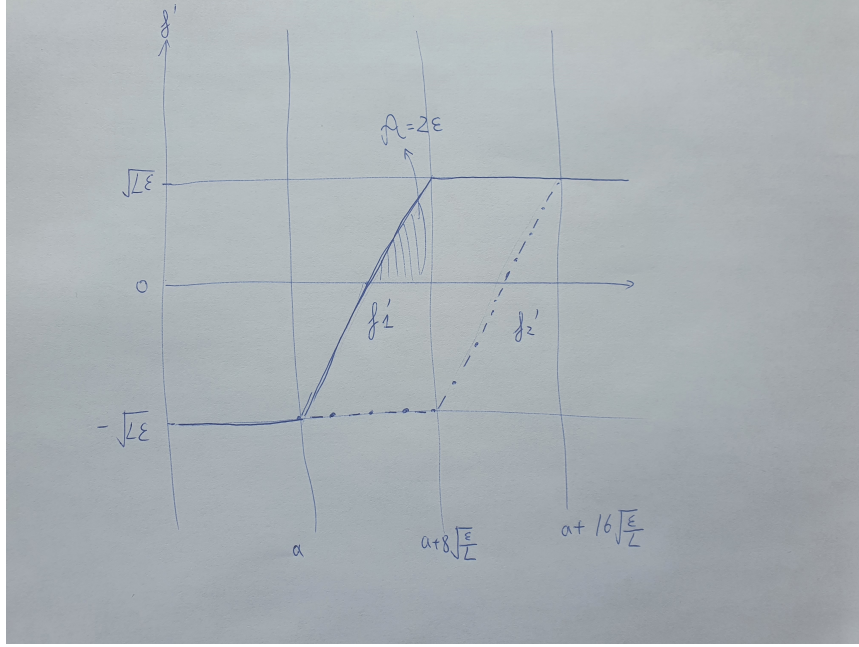


Figure 4: (f'_1, f'_2) for smooth convex lower bound

For the lower bound, we suppose we have a segment $\left[a, a + 16\sqrt{\frac{\epsilon}{L}}\right]$ such that

$$\begin{aligned} \{x_1, \dots, x_N\} \cap \left[a, a + 16\sqrt{\frac{\epsilon}{L}}\right] &= \emptyset \\ \forall i, \quad x_i \leq a &\implies f'(x_i) = -\sqrt{L\epsilon} \\ \forall i, \quad x_i \geq a + 16\sqrt{\frac{\epsilon}{L}} &\implies f'(x_i) = \sqrt{L\epsilon} \end{aligned} \tag{1}$$

We will build (f'_1, f'_2) using linear interpolation.

$$f_1'(0) = f_1'(a) = -\sqrt{L\epsilon}, \quad f_1'\left(a + 8\sqrt{\frac{\epsilon}{L}}\right) = f_1'(1) = \sqrt{L\epsilon},$$

$$f_2'(0) = f_2'\left(a + 8\sqrt{\frac{\epsilon}{L}}\right) = -\sqrt{L\epsilon}, \quad f_2'\left(a + 16\sqrt{\frac{\epsilon}{L}}\right) = f_2'(1) = \sqrt{L\epsilon}$$

We notice that :

$$\arg \min f_1 = a + 4\sqrt{\frac{\epsilon}{L}}$$

$$\arg \min f_2 = a + 12\sqrt{\frac{\epsilon}{L}}$$

So for $x \geq a + 16\sqrt{\frac{\epsilon}{L}}$,

$$f_1(x) - f_1(x_1^*) \geq \int_{a+4\sqrt{\frac{\epsilon}{L}}}^{a+8\sqrt{\frac{\epsilon}{L}}} f_1'(u) du = \frac{\sqrt{L\epsilon}}{2} \times 4\sqrt{\frac{\epsilon}{L}} = 2\epsilon > \epsilon$$

By symmetry, we can verify the same for $x \leq a$, then also similar case for f_2 .

And the smoothness of f_1, f_2 holds if :

$$\frac{2\sqrt{L\epsilon}}{8\sqrt{\frac{\epsilon}{L}}} = \frac{L}{4} \leq L$$

Now, we claim that by induction, the lower bound algorithm can satisfy the condition 1 if $N \leq \log_2 \left(\frac{1}{16} \sqrt{\frac{L}{\epsilon}} \right) = \frac{1}{2\log(2)} \log \left(\frac{L}{\epsilon} \right) - 4$, which correspond to our lower bound.

□

2.2 High dimension : Gradient descent

Now we consider $F : \mathbb{R}^d \rightarrow \mathbb{R}$, with d possibly bigger than N the evaluations of gradients allowed.

Failure of Zero order methods in high dimension

Consider $(x_1, \dots, x_N)$ to be requested and x_{out} the output, it is possible to define an orthogonal base $e_1, \dots, e_{N+1}$ such that :

$$x_i \in \text{Span}(e_1, \dots, e_{N+1})$$

then one could extend the base and define $F(X) = f(\langle X, e_{N+2} \rangle)$, the solver only gets $f(0)$ and therefore cannot be close to the minimum of f .

Gradient descent

Lemma 2.2. *If F is convex, ∇F is L -Lipschitz, it is possible to find an ϵ -close solution with $O\left(\frac{L\|x-x_\star\|^2}{\epsilon}\right)$ requests of the gradient ∇F .*

Proof. We define the gradient descent as :

$$X_{t+1} = X_t - \eta_t \nabla F(x_t)$$

We remind by convexity that $F(X_t) - F(X_\star) \leq \langle \nabla F(X_t), X_t - X_\star \rangle$, therefore :

$$\begin{aligned} & \|X_{t+1} - X_\star\|^2 \\ &= \|X_t - X_\star\|^2 - 2\eta_t \langle \nabla F(X_t), X_t - X_\star \rangle + \eta_t^2 \|\nabla F(x_t)\|^2 \end{aligned}$$

then :

$$\begin{aligned} & 2\eta_t (F(X_t) - F(X_\star)) \\ & \leq 2\eta_t \langle \nabla F(X_t), X_t - X_\star \rangle \\ &= \|X_t - X_\star\|^2 - \|X_{t+1} - X_\star\|^2 + \eta_t^2 \|\nabla F(x_t)\|^2 \end{aligned}$$

We notice by smoothness that :

$$\begin{aligned} D^F(X_{t+1}, X_t) &= F(X_{t+1}) - F(X_t) - \langle \nabla F(X_t), X_{t+1} - X_t \rangle \\ &\leq \frac{L}{2} \|X_{t+1} - X_t\|^2 \\ F(X_{t+1}) - F(X_t) &\leq \left(\frac{L\eta_t^2}{2} - \eta_t \right) \|\nabla F(x_t)\|^2 \end{aligned}$$

By convenience if we choose $\eta_t = \frac{1}{L}$,

$$F(X_t) - F(X_{t+1}) \geq \frac{1}{2L} \|\nabla F(x_t)\|^2$$

And combine with the previous equation :

$$\begin{aligned} & \frac{2}{L} (F(X_t) - F(X_\star)) \\ & \leq \|X_t - X_\star\|^2 - \|X_{t+1} - X_\star\|^2 + \frac{1}{L^2} \|\nabla F(x_t)\|^2 \\ & \leq \|X_t - X_\star\|^2 - \|X_{t+1} - X_\star\|^2 + \frac{2}{L} (F(X_t) - F(X_{t+1})) \end{aligned}$$

So we only need to use the telescopic sum from 1 to N :

$$\begin{aligned} & \frac{2}{L} (F(X_{t+1}) - F(X_\star)) \leq \|X_t - X_\star\|^2 - \|X_{t+1} - X_\star\|^2 \\ \frac{2N}{L} (F(X_N^{ag}) - F(X_\star)) & \leq \frac{2}{L} \sum_{t=1}^N (F(X_{t+1}) - F(X_\star)) \leq \|X_1 - X_\star\|^2 - \|X_{N+1} - X_\star\|^2 \end{aligned}$$

the last line is to define $X_N^{ag} = \frac{1}{N} \sum_{t=1}^N X_{t+1}$ and use convexity. Then we obtain that :

$$F(X_N^{ag}) - F(X_*) \leq \frac{L\|X_1 - X_*\|^2}{2N} = \epsilon$$

A quick manipulation brings us to $N = \frac{L\|X_1 - X_*\|^2}{2\epsilon}$.

□

Unlike the previous cases, this time the simple method does not bring us to the optimal performance. In fact, we can use accelerated gradient descent in that case and obtain better performance, which we can also prove to be optimal.

Lemma 2.3. *If F is convex, ∇F is L -Lipschitz and $d \geq 2N$, the number of gradients require ∇F to find an ϵ -close solution is $\Omega\left(\sqrt{\frac{L\|x - x_*\|^2}{\epsilon}}\right)$.*

To sum up :

Conditions	d=1	d » N
Convexity + Zero order method	✓	✗
Convexity + First order method	✓	✓

Table 1: Solvable convex problems